

Chapter I: Introduction to Sampled Data Systems

Automation and Control Engineering

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1 Introduction

The fundamental difference between a traditional analog control loop and a digital control system lies in the presence of sampling and digital processing elements. Figure 1 compares both architectures.

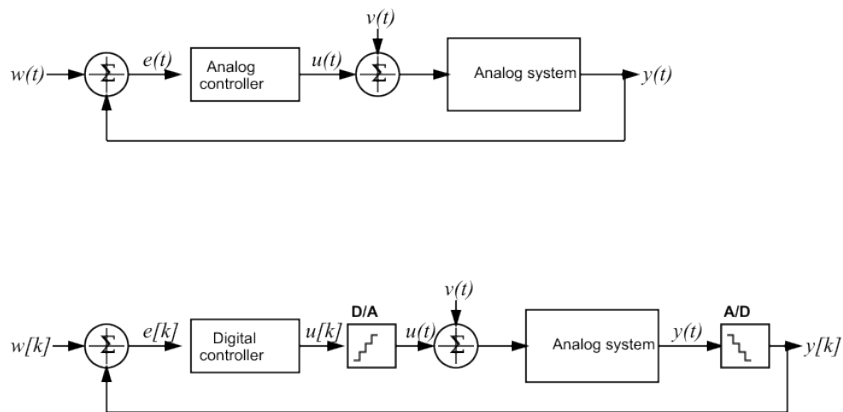


Figure 1: Comparison between analog and digital control systems

1.1 System Components

- $v(t)$: Disturbance signal
- $w(t), w(k)$: Analog and sampled reference signals
- $y(t), y(k)$: Analog and sampled output signals
- A/D: Analog-to-Digital Converter
- D/A: Digital-to-Analog Converter

1.2 Advantages of Digital Control Systems

- **Flexibility:** Easy implementation of complex algorithms
- **Resource Sharing:** A single processor can control several loops
- **Data Logging:** Capability to record and analyze long-term data
- **Adaptability:** Parameters can be adjusted in real time
- **User Interface:** Integration with graphical tools and monitoring platforms

1.3 Main Challenge

- **Information Loss:** Sampling may cause aliasing and performance degradation

2 Sampling and Quantization

2.1 Sampling Process

The A/D converter transforms the continuous signal $y(t)$ into discrete values $y(kT_s)$ taken at sampling instants defined by the sampling period T_s :

$$[y(0), y(T_s), y(2T_s), y(3T_s), \dots, y(kT_s), \dots]$$

For constant sampling intervals, this is simplified to:

$$[y(0), y(1), y(2), y(3), \dots, y(k), \dots]$$

2.2 Quantization

The A/D converter delivers digital values, typically 8 or 16 bits, representing quantized samples of the analog signal. The resulting signal is both discrete in time and quantized in amplitude.

3 Determination of the Sampling Period

The sampling period T_s is a key design parameter directly influencing system stability and performance.

3.1 Nyquist–Shannon Sampling Theorem

A continuous-time signal $x(t)$ with maximum frequency f_{max} can be perfectly reconstructed from its samples if it is sampled at a frequency:

$$f_s > 2f_{max}$$

3.2 Practical Considerations

- Small T_s : High precision but heavy computational load
- Large T_s : Possible signal distortion and control delay
- Industrial rule: $f_s = 5\text{--}10\times$ system bandwidth

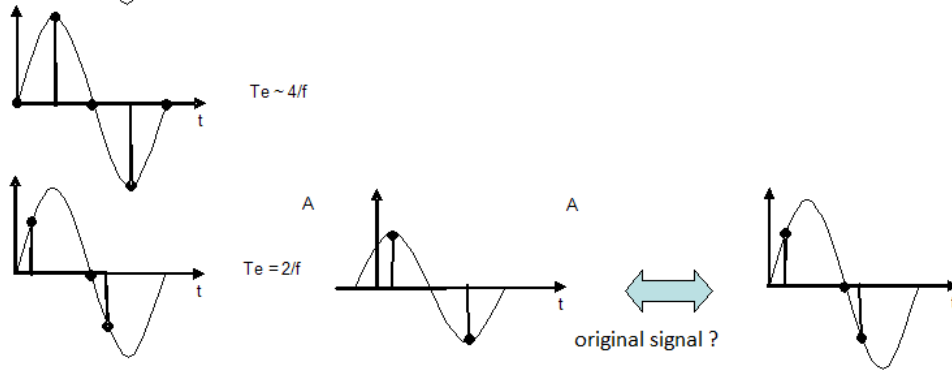


Figure 2: Sampling period selection based on Shannon's theorem

4 Sampling Period for Continuous Systems

When discretizing a continuous system, the choice of the sampling period (or frequency) directly affects the closed-loop performance. A widely accepted empirical rule states:

For continuous systems, the sampling frequency f_e is generally chosen between 6 and 24 times the closed-loop bandwidth frequency $f_{BP,CL}$.

Depending on the transfer function structure, the following guidelines can be applied:

4.1 Sampling Period of a First-Order System

For a first-order system with transfer function

$$G(s) = \frac{k}{1 + \tau s},$$

the sampling period T_e should be selected based on the system time constant τ . To ensure accurate digital representation of the continuous dynamics, the following practical rule is applied:

$$\tau \geq T_e \geq \frac{\tau}{4}.$$

This range ensures that the sampling frequency is sufficiently high compared to the dominant time constant of the system.

4.2 Sampling Period of a Second-Order System

For a second-order system, the sampling period T_e should be chosen according to the system's natural frequency ω_0 and damping ratio ζ . Depending on the damping ratio, the following practical ranges apply:

$$0.25 \leq \omega_0 T_e \leq 1; \quad \zeta = 0.7$$

$$0.4 \leq \omega_0 T_e \leq 1.75; \quad \zeta = 1$$

A general range that provides a good trade-off between performance and numerical stability is:

$$\boxed{0.25 \leq \omega_0 T_e \leq 1.5; \quad 0.7 \leq \zeta \leq 1}$$

This choice guarantees a sufficiently fast sampling rate to capture the dynamic behavior without excessive computational load.

4.3 System of Order n

For:

$$G(s) = \frac{K}{(1 + \tau_1 s)(1 + \tau_2 s) \cdots (1 + \tau_n s)},$$

a practical range is:

$$\frac{\tau_{\min}}{10} < T_e < \frac{\tau_{\min}}{5},$$

where $\tau_{\min} = \min(\tau_i)$.

5 Reconstruction of the Analog Signal

The digital controller outputs a discrete command $u(k)$, which must be converted back into a continuous signal $u(t)$ to drive the analog plant.

5.1 Zero-Order Hold (ZOH)

The ZOH maintains the last sample value constant during the entire sampling period:

$$u_a(t) = u[k], \quad \text{for } kT_s \leq t < (k+1)T_s$$

5.2 Higher-Order Holds

Higher-order holds (e.g., first-order holds using linear extrapolation) generate smoother signals but increase computational complexity and delay.

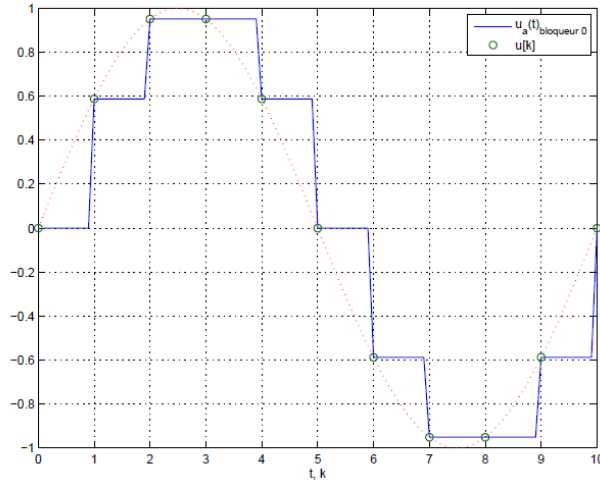


Figure 3: Signal reconstruction using zero-order hold

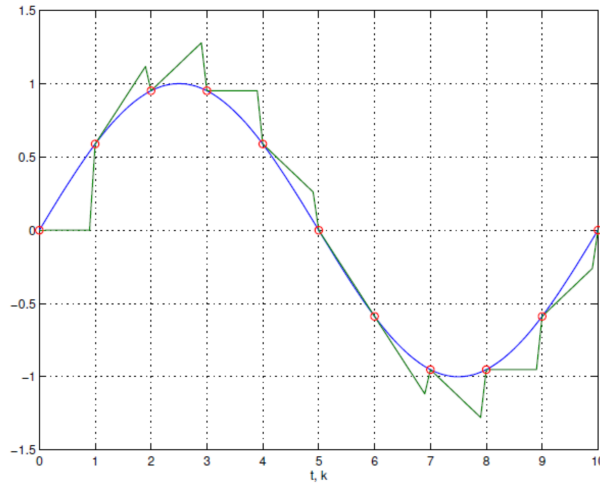


Figure 4: First-order hold reconstruction

6 Mathematical Representation of Sampling

The sampling operation can be modeled as:

$$x_s(t) = x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

where $\delta(t)$ denotes the Dirac delta function. This formalism allows frequency-domain analysis using Fourier transforms and the concept of spectral replication.

7 References

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